

A supercritical exponential-Orlicz lower gain for discrepancy

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Abstract

Bilyk and Markhasin ask for lower bounds matching their exponential-Orlicz upper estimates for discrepancy in dimensions $d \geq 3$. The critical $\exp(L^{2/(d-1)})$ lower bound is already Roth's L_2 bound. We extract additional information from the short Riesz-product proof of Bilyk–Lacey–Vagharshakyan: their L_1 test function has enough L_2 control to give a growing- p lower bound. Consequently, for every $\beta > 2$, the $\exp(L^\beta)$ discrepancy of every N -point set is larger than the Roth order by a positive power of $\log N$ (up to one $\log \log N$ factor). This is a partial result; it does not reach the conjectural matching exponent.

1 Statement

For an N -point set $\mathcal{P}_N \subset [0, 1)^d$, use the unnormalised anchored discrepancy

$$D_{\mathcal{P}_N}(x) = \sum_{z \in \mathcal{P}_N} \mathbf{1}_{[0, x)}(z) - Nx_1 \cdots x_d.$$

As usual, $\|f\|_{\exp(L^\beta)}$ denotes the Luxemburg norm associated to a Young function equal to $e^{t^\beta} - 1$ for large t . On a probability space,

$$\|f\|_{\exp(L^\beta)} \asymp_\beta \sup_{p > 1} p^{-1/\beta} \|f\|_{L^p}. \quad (1)$$

Theorem 1 (supercritical Orlicz gain). *For every $d \geq 3$ there is $\eta_d > 0$ such that, for every $\beta > 2$, there is $c_{d, \beta} > 0$ with the following property. Every N -point set $\mathcal{P}_N \subset [0, 1)^d$, $N \geq 3$, satisfies*

$$\|D_{\mathcal{P}_N}\|_{\exp(L^\beta)} \geq c_{d, \beta} (\log N)^{\frac{d-1}{2} + \eta_d(1-2/\beta)} (\log \log(e^e N))^{-1/\beta}. \quad (2)$$

One may take $\eta_d = \varepsilon_d/4$, where $\varepsilon_d > 0$ is any admissible short-product exponent in the proof of Bilyk–Lacey–Vagharshakyan's improved star-discrepancy lower bound.

For every fixed $\beta > 2$, the exponent in (2) is strictly larger than Roth's $(d-1)/2$. As $\beta \rightarrow \infty$, it tends to the star-discrepancy gain $(d-1)/2 + \eta_d$ obtained by the same short Riesz product.

2 The growing-moment extraction

Fix n by $2N \leq 2^n < 4N$. We recall only the quantitative properties of the test function constructed in Section 9 of Bilyk–Lacey–Vagharshakyan (BLV). Choose a sufficiently small $\varepsilon = \varepsilon_d > 0$, put

$$q = \lfloor an^\varepsilon \rfloor, \quad b = \frac{1}{4}, \quad \tilde{\rho} = aq^b n^{-(d-1)/2},$$

and use the separated blocks of multiindices in the discrepancy part of their proof. The blocks have length comparable to n/q ; this is also the length used explicitly in their estimates following the displayed block definition. For the BLV r -functions $f_{\mathbf{r}}$ let $F_t = \sum_{\mathbf{r} \in \mathbb{A}_t} f_{\mathbf{r}}$ and form the short product

$$\Psi = \prod_{t=1}^q (1 + \tilde{\rho} F_t) = 1 + \Psi^{\text{sd}} + \Psi^{\neg}.$$

Here Ψ^{sd} retains the products whose multiindices are strongly distinct and $\Psi^{\neg}$ is the coincidence remainder.

The BLV proof gives

$$\left\| \Psi^{\text{sd}} \right\|_{L^1} \leq C_d, \quad (3)$$

$$\langle D_{\mathcal{P}_N}, \Psi^{\text{sd}} \rangle \geq c_d q^{1/4} n^{(d-1)/2}. \quad (4)$$

Indeed, the linear term has the size on the right of (4), whereas all higher strongly-distinct terms pair with discrepancy by a summable amount. We next record the extra estimate that is not extracted in BLV.

Proposition 1 (near- L^1 control of the test). *For q sufficiently large,*

$$\left\| \Psi^{\text{sd}} \right\|_{L^2} \leq \exp(C_d q^{1/2} \log(2q)). \quad (5)$$

Consequently, if $p \geq C_d q^{1/2} \log(2q)$ and $p' = p/(p-1)$, then

$$\left\| \Psi^{\text{sd}} \right\|_{L^{p'}} \leq C_d. \quad (6)$$

Proof. The proof uses estimates already established inside the BLV short-product argument. Their Lemma 5.3, uniformly for partial products, gives

$$\left\| \prod_{t \in V} (1 + \tilde{\rho} F_t) \right\|_2 \leq \exp(C_d q^{2b}) = \exp(C_d q^{1/2}). \quad (7)$$

Their proof also records the crude fourth-moment estimate

$$\left\| \prod_{t \in V} (1 + \tilde{\rho} F_t) \right\|_4 \leq (C_d q)^q, \quad (8)$$

again uniformly in $V \subset \{1, \dots, q\}$.

Write the inclusion–exclusion expansion of the coincidence remainder in the form

$$\Psi^{\neg} = \sum_G A_G B_G,$$

where G runs over the admissible coincidence graphs, $A_G = \tilde{\rho}^{|V(G)|} \text{SumProd}(X(G))$, and B_G is a partial product over the unused blocks. The estimates in BLV that prove $\|\Psi^{\neg}\|_{L^1} \lesssim 1$ actually establish the stronger summability statement

$$\sum_G \|A_G\|_{L^s} \leq C_d, \quad s = q^{2b} = q^{1/2}. \quad (9)$$

For $s \geq 4$, choose r by $1/2 = 1/s + 1/r$. Thus $r = 2s/(s-2) \in [2, 4]$. Interpolating (7) and (8), with interpolation parameter $4/s$, yields

$$\sup_V \left\| \prod_{t \in V} (1 + \tilde{\rho} F_t) \right\|_r \leq \exp(C_d q^{1/2} \log(2q)). \quad (10)$$

Hölder, (9), and (10) now imply

$$\|\Psi^\neg\|_{L^2} \leq \sum_G \|A_G\|_{L^s} \|B_G\|_{L^r} \leq \exp(C_d q^{1/2} \log(2q)).$$

Since $\Psi^{\text{sd}} = \Psi - 1 - \Psi^\neg$ and (7) also bounds $\|\Psi\|_{L^2}$, this proves (5).

Finally, interpolate (3) and (5). Since

$$\frac{1}{p'} = 1 - \frac{1}{p} = (1 - \theta) \cdot 1 + \theta \cdot \frac{1}{2}, \quad \theta = \frac{2}{p},$$

we get

$$\|\Psi^{\text{sd}}\|_{L^{p'}} \leq C_d \exp\left(\frac{C_d q^{1/2} \log(2q)}{p}\right).$$

This is bounded when $p \geq C_d q^{1/2} \log(2q)$. □

Proposition 2 (growing- p discrepancy lower bound). *With the notation above,*

$$\|D_{\mathcal{P}_N}\|_{L^p} \geq c_d q^{1/4} n^{(d-1)/2} \quad \text{whenever } p \geq C_d q^{1/2} \log(2q). \quad (11)$$

Proof. Apply Hölder to (4) and use (6). □

3 Proof of Theorem 1

Take p comparable to $q^{1/2} \log(2q)$ in (1). Proposition 2 gives

$$\begin{aligned} \|D_{\mathcal{P}_N}\|_{\exp(L^\beta)} &\gtrsim_{d,\beta} p^{-1/\beta} q^{1/4} n^{(d-1)/2} \\ &\gtrsim_{d,\beta} n^{(d-1)/2} q^{1/4-1/(2\beta)} (\log(2q))^{-1/\beta}. \end{aligned}$$

Since $q \asymp n^{\varepsilon_d}$ and $\eta_d = \varepsilon_d/4$, this is

$$n^{(d-1)/2 + \eta_d(1-2/\beta)} (\log(2n))^{-1/\beta}.$$

Finally $n \asymp \log N$. Enlarging the constant handles the finitely many small values for which $q < 4$, and proves (2).

4 Scope and obstruction to a full result

The Bilyk–Markhasin upper bound for $\beta \geq 2/(d-1)$ has order $(\log N)^{d-1-1/\beta}$. Theorem 1 is not matching. At the critical exponent $2/(d-1)$, the Roth lower bound already gives the correct order of their critical upper theorem; the open difficulty concerns the stronger supercritical scale.

The threshold $\beta > 2$ is intrinsic to the present extraction. If one keeps a general short-product parameter $b \leq 1/4$, positivity and the second-moment argument permit a gain q^b , while controlling the coincidence remainder in L^2 through the available graph estimate costs moments at scale q^{1-2b} (up to logarithms). The inequality $b > (1-2b)/\beta$ is easiest at the extremal admissible value $b = 1/4$ and then becomes exactly $\beta > 2$. Reaching $\beta \leq 2$, much less the full matching scale, requires a genuinely stronger coincidence estimate or a different test function. Such an improvement is tied to the unresolved higher-dimensional small-ball/star-discrepancy problem.

References

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